RELOCALIZING A LOCALIZATION OF A TOPOS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The localization of a topos in a slice. Same chapter, Lemma lemma-localize-topos. Supplies the descriptions of the pullback and the lower shriek of the localization.

Source reference. Original lemma 04GZ: sources/stacks-sites/pinned-original-source.tex, source lines 7721–7721. Navigation pointer invoked at proof line 7721; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. If $s : \mathcal{G} \to \mathcal{F}$ is a morphism of sheaves on $\mathcal{C}$ then there exists a natural commutative diagram of morphisms of topoi*

$$\begin{array}{ccc} Sh(\mathcal{C})/\mathcal{G} & \xrightarrow{\quad j \quad} & Sh(\mathcal{C})/\mathcal{F} \\ & {\scriptstyle j_{\mathcal{G}}} \searrow \quad \swarrow {\scriptstyle j_{\mathcal{F}}} & \\ & Sh(\mathcal{C}) & \end{array}$$

where $j = j_{\mathcal{G}/\mathcal{F}}$ is the localization of the topos $Sh(\mathcal{C})/\mathcal{F}$ at the object $\mathcal{G}/\mathcal{F}$. In particular we have

$$j^{-1}(\mathcal{H} \to \mathcal{F}) = (\mathcal{H} \times_{\mathcal{F}} \mathcal{G} \to \mathcal{G})$$

and

$$j_!(\mathcal{E} \xrightarrow{e} \mathcal{G}) = (\mathcal{E} \xrightarrow{s \circ e} \mathcal{F}).$$

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Proof. The description of j^{-1} and $j_!$ comes from the description of those functors in Lemma 04GZ. The equality of functors $j_{\mathcal{G}!} = j_{\mathcal{F}!} \circ j_!$ is clear from the description of these functors (as forgetful functors). By adjointness we also obtain the equalities $j_{\mathcal{G}}^{-1} = j^{-1} \circ j_{\mathcal{F}}^{-1}$, and $j_{\mathcal{G},*} = j_{\mathcal{F},*} \circ j_*$. □